

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>
Melanie has 5 vacuum cleaners left, which are half of what was left at the orange house, so she had:

$$\backslash[5 \times 2 = 10 \text{ vacuum cleaners at the orange house} \backslash]$$

Before she went to the orange house, she had:

$$\backslash[10 \times 2 = 20 \text{ vacuum cleaners} \backslash]$$

She sold a third of her vacuum cleaners at the green house, so she had:

$$\backslash[20 - \frac{1}{3} \times 20 = \frac{2}{3} \times 20 = \frac{40}{3} \text{ vacuum cleaners left} \backslash]$$

Then, she sold 2 more to the red house, so she had:

$$\backslash[\frac{40}{3} - 2 = \frac{40}{3} - \frac{6}{3} = \frac{34}{3} \text{ vacuum cleaners left} \backslash]$$

Thus, the number of vacuum cleaners Melanie started with is:

$$\backslash[\frac{34}{3} \times 3 = 34 \text{ vacuum cleaners} \backslash]</think>$$

<answer>34</answer>